

```

GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define SERVER_FIFO "/tmp/os_server_fifo"
#define BUFFER_SIZE 256

int main(void) {
    char buffer[BUFFER_SIZE];
    int server_fd, client_fd;
    ssize_t n;

    unlink(SERVER_FIFO);

    if (mkfifo(SERVER_FIFO, 0666) == -1) {
        perror("mkfifo");
        return 1;
    }

    printf("Server started. Waiting for client messages...\n");

    server_fd = open(SERVER_FIFO, O_RDONLY);
    if (server_fd == -1) {
        perror("open server FIFO");
        return 1;
    }

    while (1) {
        memset(buffer, 0, sizeof(buffer));

        n = read(server_fd, buffer, sizeof(buffer) - 1);

        if (n > 0) {
            buffer[n] = '\0';

            char *separator = strchr(buffer, '|');

            if (separator != NULL) {
                *separator = '\0';

                char *client_fifo = buffer;
                char *message = separator + 1;

                printf("Received: %s\n", message);

                client_fd = open(client_fifo, O_WRONLY);

                if (client_fd == -1) {
                    perror("open client FIFO");
                    continue;
                }

                const char *response =
                    "Message received successfully";

                write(client_fd, response, strlen(response) + 1);
                close(client_fd);

                if (strcmp(message, "exit") == 0)
                    break;
            }
        } else if (n == 0) {
            close(server_fd);
            server_fd = open(SERVER_FIFO, O_RDONLY);
        } else {
            perror("read");
            break;
        }
    }

    close(server_fd);
    unlink(SERVER_FIFO);

    printf("Server stopped. FIFO removed.\n");

    return 0;
}

```

```

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#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>

#define SERVER_FIFO "/tmp/os_server_fifo"
#define BUFFER_SIZE 256

int main(void) {
    char message[BUFFER_SIZE];
    char response[BUFFER_SIZE];
    char client_fifo[BUFFER_SIZE];

    int server_fd, client_fd;
    ssize_t n;

    snprintf(client_fifo, sizeof(client_fifo),
             "/tmp/client_fifo_%d", getpid());

    unlink(client_fifo);

    if (mkfifo(client_fifo, 0666) == -1) {
        perror("mkfifo");
        return 1;
    }

    printf("Enter message: ");
    fgets(message, sizeof(message), stdin);

    message[strcspn(message, "\n")] = '\0';

    server_fd = open(SERVER_FIFO, O_WRONLY);

    if (server_fd == -1) {
        perror("open server FIFO");
        unlink(client_fifo);
        return 1;
    }

    char request[BUFFER_SIZE * 2];

    snprintf(request, sizeof(request),
             "%s%s", client_fifo, message);

    write(server_fd, request, strlen(request) + 1);
    close(server_fd);

    client_fd = open(client_fifo, O_RDONLY);

    if (client_fd == -1) {
        perror("open client FIFO");
        unlink(client_fifo);
        return 1;
    }

    n = read(client_fd, response, sizeof(response) - 1);

    if (n > 0) {
        response[n] = '\0';
        printf("Server response: %s\n", response);
    }
}

```

Output

```

Last login: Thu Sep 10 10:30:12 2026 from 192.168.252.1
ubuntu@ubuntu:~$ cd OSLab
ubuntu@ubuntu:~/OSLab$ ls
FileDescriptors Practical_Practical_2 Skill exec fork program singal
ubuntu@ubuntu:~/OSLab$ cd Practical
ubuntu@ubuntu:~/OSLab/Practical$ ls
2 fifo_client.c fifo_server.c practical6 program2.c program3.c program4.c program5.c program_1.c program_2- source
ubuntu@ubuntu:~/OSLab/Practical$ ./fifo_client
-bash: ./: Is a directory
ubuntu@ubuntu:~/OSLab/Practical$ ./fifo_client
Enter message: Hello Server
Server response: Message received successfully
ubuntu@ubuntu:~/OSLab/Practical$

```